

Jupiter

Overview

Jupiter is the fifth planet from the Sun and by far the largest in the Solar System, with a mass of 1.898×10^{27} kg — more than twice the combined mass of all other planets, and 318 times the mass of Earth. Its equatorial diameter of 142,984 km is roughly 11 times that of Earth, and its volume is so large that more than 1,300 Earths could fit inside. Jupiter is a gas giant, meaning it has no solid surface; its atmosphere transitions gradually from gaseous hydrogen and helium at the top to increasingly dense and hot fluid deeper inside. Jupiter completes one rotation in just under 10 hours, making it the fastest-rotating planet in the Solar System; this rapid spin causes the planet to bulge at the equator and flatten at the poles, giving it an oblate spheroid shape with an equatorial diameter about 9,000 km larger than its polar diameter.

Atmosphere and Storms

Jupiter's atmosphere is divided into alternating light zones and dark belts that encircle the planet parallel to its equator, driven by powerful jet streams reaching speeds of 500–600 km/h. The light zones are regions of rising gas, while the dark belts are regions of sinking gas, creating a complex system of counter-rotating wind bands. The most famous feature is the Great Red Spot, an anticyclonic high-pressure storm larger than Earth that has been continuously observed since at least 1831, though historical records suggest it may have existed for over 350 years. Since the late 19th century, the Great Red Spot has been shrinking; it was once three times Earth's diameter but is now barely larger than Earth. Jupiter also hosts numerous smaller storms, including Oval BA — a storm the size of Earth that formed from the merger of three smaller storms in 2000 and turned red in 2006, earning it the nickname 'Red Spot Jr.' Lightning storms on Jupiter are estimated to be up to three times as energetic as those on Earth and have been detected at both poles by the Juno spacecraft.

Interior Structure and Magnetosphere

Below Jupiter's visible cloud tops, the pressure and temperature increase rapidly with depth. At around 20,000 km below the cloud tops, the pressure becomes so extreme that hydrogen gas transitions into a metallic state — liquid metallic hydrogen — which conducts electricity and flows freely. This vast ocean of metallic hydrogen, combined with Jupiter's rapid rotation, generates the strongest magnetic field of any planet in the Solar System — about 20,000 times stronger than Earth's. Jupiter's magnetosphere is the largest structure in the Solar System aside from the heliosphere itself; if visible to the human eye, it would appear several times larger than the full Moon in the night sky. The intense radiation belts trapped in Jupiter's magnetosphere are so powerful that they would deliver a lethal dose of radiation to an unshielded human in minutes, and even damage spacecraft electronics. At the core, there is thought to be a dense, fuzzy region of rock, metal, and compressed fluid roughly 10–15 times the mass of Earth, though its exact nature is still being studied by the Juno mission.

The Galilean Moons

Jupiter's four largest moons — Io, Europa, Ganymede, and Callisto — were discovered by Galileo Galilei in January 1610 and are known collectively as the Galilean moons. Io is the most volcanically active body in the Solar System, with over 400 active volcanoes driven by tidal heating as Jupiter's gravity stretches and squeezes the moon; some volcanic plumes reach 500 km into space. Europa has a smooth icy surface beneath which lies a global saltwater ocean estimated to contain twice the volume of all of Earth's oceans combined, making it one of the most promising candidates for extraterrestrial life. Ganymede is the largest moon in the Solar System, bigger than the planet Mercury, and is the only moon known to have its own magnetic field. Callisto is the most heavily cratered object in the Solar System and has a very ancient, unchanged surface; evidence suggests it too may have a subsurface ocean of liquid water. NASA's Europa Clipper mission, launched in October 2024, will conduct detailed flybys of Europa to assess its habitability.